

Graph-aware VCF

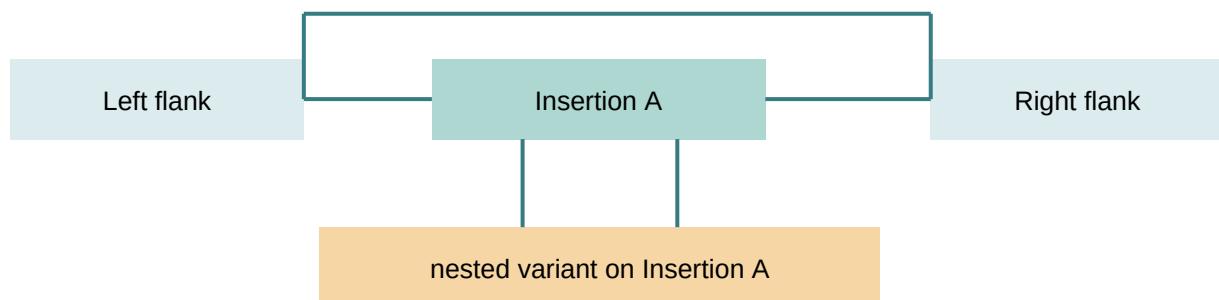
Lossless assemblies through nested variants

LinGraph is a variants-guided **pangenome graph builder**. Its grVCF format is designed primarily for **lossless representation of assembly sequences**, with an optional exact representation of variants.

Two purposes, one hierarchy

- 1. Lossless assembly representation.** Preserve the information needed to reproduce each haplotype sequence across its represented regions.
- 2. Unambiguous variants (optional exact merge).** Every carrier of a line shares the same allele sequence and reference coordinates in that line's coordinate space.

A **hierarchy of nested variants** makes both possible. Similar alleles share a parent; differences are child variants on that parent, with further children as needed. Each line can describe one exact operation while different parent-plus-child combinations reconstruct different assemblies.



Insertion A adds an alternative route between reference flanks. The nested variant changes sequence within A. Worked examples are on page 5.

Keep the complete representation

Retain all variant classes, parent and child records, placement metadata, and referenced sequence catalogs. SV-only output cannot represent every assembly difference. Lossless reconstruction concerns represented regions; unmapped assembly bases need separate accounting.

Graph export is a downstream use: grVCF can become a pangenome graph. Standard VCF export removes nested variation and is not lossless.

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These are LinGraph conventions, not a separate standardized VCF version. The current rGFA exporter includes all variant sizes by default.

01 / Read a record

grVCF uses a VCF 4.2-style envelope. INFO defines the row allele; FORMAT records its carriers and assembly coordinates. In exact output, a line has one sequence and one reference placement. Apply its carrier-specific descendants to reconstruct the full assembly allele. [1, 2, 3]

Column / concept	How to read it
CHROM	A reference/alternative locus or a parent variant ID. If it is an insertion ID, POS refers to that inserted sequence, not to the original chromosome.
POS / END	Event coordinates. SNP base positions and structural-variant breakpoints follow different conventions; see page 4.
ID	A stable variant identifier that may also name a sequence target and parent coordinate space. Keep dependencies when subsetting or renaming.
REF / ALT	SNPs have base alleles; structural events may use <INS>, , or <SUB>. Representative query sequence can be stored in INFO/SEQ.
QUAL / FILTER	Preserve emitted values. Export filters and required dependencies determine which branches are retained.
INFO / FORMAT	Exact mode: one shared allele and placement; differences are separate/nested records. Non-exact mode: FORMAT can retain differences from the representative.
Sample columns	Usually assembly haplotypes such as HG002_h1 and HG002_h2. These are not sequencing-depth measurements.

Genotypes and multiple observations

GT=1 means the haplotype supports the variant; **GT=0** is callable reference; **GT=.** is no call. Never turn missing data into reference support. Merging mode and available mapping coverage determine reference calls.

Read each FORMAT name from column 9, then split sample cells on colons. Number=. observation fields contain comma-separated values; preserve correspondence across those lists. GT remains one state. Decode escaped characters only after splitting the structural delimiters.

Exact and non-exact merging

Exact mode: carriers on a row share its sequence and reference coordinates. A shifted observation is realigned; it joins that placement only when it can be represented exactly, otherwise it is separated. Differences within a parent allele become child lines, recursively.

Non-exact mode: a row can group observations with different sequences, sizes, or breakpoints. FORMAT retains member information, including TEMPLATEOFFSET. A nonzero offset belongs to this representation, not an exact shared placement.

The cohort wrapper defaults to exact merging when no explicit variant-selection mode is given. Direct graphvcfmerge.py usage enables it with --exact QUERY_PATHS and --reference FASTA; retain indexed assemblies and all required references. [3]

02 / INFO fields

Number=1 unless noted. Field availability depends on the producing stage; the file header is the schema for that file. [2, 3]

Field	Type	Meaning
SVTYPE	String	Event type: INS, DEL, SUB, or SNP. Merging may decompose replacement events into components.
END	Integer	Event end/second anchor. A merged deletion can carry a cluster envelope; use representative SVLEN to recover its actual deletion span.
NSUP	Integer	Number of supporting sample/haplotype columns, not read depth.
SVLEN	Integer	Representative signed length: insertion positive, deletion negative.
MAXSIZE	Integer	Maximum insertion/deletion component size among cluster members, not necessarily the net length change.
SEQ	String	Representative query allele. Default final caller output is plain sequence; merged insertions may use named graph-encoded pieces. Dot means absent.
EXTENDGRAPHCIGAR	String	Representative extended graph alignment. Default final caller output removes inline sequence payloads.
PACCLASS	String	Representative pseudo-linear assignment class: primary or fulllocusdup.
PAMAP	Integer; Number=.	IDs of originating <code>##pseudoLinearMapping</code> entries. Interpret their Path values to distinguish location from sequence-source information.
ALTERNATIVECIGAR	String	Query-to-source-locus alignment for a full-locus duplication insertion; adjacent elected pieces are joined with &.

Location and sequence source are different

A `##pseudoLinearMapping` entry contains ID, Sample, Query, Reference, Category, and Path. Its coordinates are declared **0-based half-open**. **Path=pri** locates the haplotype bases; **Path=alt** describes a sequence-source layer and adds no extra bases to the haplotype. INFO/PAMAP cites these entries in caller output. Do not count an alt-source mapping as another copy of the haplotype. [2]

Mapping IDs are scoped to their source header. Merged output need not retain the caller mapping layer: current merging filters alt-only records and removes mapping-specific metadata. Keep original sample VCFs for assembly-walk reconstruction. [3, 6]

Older or intermediate fields

CIGAR, SVINSSEQ, SVDELSEQ, and QUERYSEQ occur in internal/older files. The final caller replaces their sequence representation with SEQ and EXTENDGRAPHCIGAR. Legacy HSV/HSNP packed sample payloads remain accepted by downstream readers.

03 / FORMAT and coordinates

FORMAT field	Number / Type	Meaning
GT	1 / String	Haploid variant/reference/no-call state.
TYPE / SIZE	. / String; Integer	Observation event type and size. Merged DEL SIZE may be positive; use TYPE for direction and INFO/SVLEN for signed representative length.
EXTENDGRAPHICIGAR	. / String	SV observation aligned against a representative or named target. Bare X operations may lack alternate bases.
BASE	. / String	Alternate base for SNP observations.
ASSEMBLYCONTIG	. / String	Source assembly contig; must match the supplied FASTA.
QUERYCOORD	. / String	0-based query coordinate or range plus strand. A minus-strand single-SNP coordinate is a traversal boundary; its base lies at coordinate minus one.
TEMPLATEOFFSET	. / Integer	Member-breakpoint displacement from row POS: positive right, negative left. Exact shared placements have offset 0; non-exact members may differ.
LIFTCATEGORY	. / String	GenomeLift category: Pri (types 1/2), Dup (type 3), DivergeDup (type 4), or .
ALLELENAME / LABEL_H	. / String	Optional PA allele/coordinate provenance. Contributors may be joined with &. Final caller drops these by default; --PAtag retains them.

Normalize coordinates before interpreting topology

Event / parent	Zero-based interval used by current rGFA exporter
SNP on any parent	[POS - 1, POS): SNP POS is a 1-based base coordinate.
INS/SUB on root	[POS, END); pure INS with END=POS has an empty replacement interval.
INS/SUB on insertion	Start = POS - 1 by default. Pure INS ends at start; nonempty replacement ends at END.
DEL	Start follows the root/nested breakpoint convention; end = start + abs(SVLEN). Only legacy rows without SVLEN use END.

These are the cohort writer/exporter conventions, not a universal interpretation of symbolic VCF. --nested-pos-base 0 is only for inputs using zero-based nested breakpoints. A D inside an insertion alignment consumes template sequence; it is not automatically a deletion on the chromosome. [4]

04 / The nested hierarchy

Four haplotypes share an insertion at chr1:100, but their final inserted sequences differ. These abbreviated teaching records show the allele definitions; the table below supplies each haplotype's carried lines. Headers and FORMAT observations are omitted.

```
CHROM POS ID REF ALT INFO
chr1 100 A N <INS> SVTYPE=INS;END=100;SVLEN=8;
      SEQ=ACGTTGCA
A 5 B N <INS> SVTYPE=INS;END=5;SVLEN=2;SEQ=GG
A 3 S G A SVTYPE=SNP
A 3 D N <DEL> SVTYPE=DEL;END=4;SVLEN=-2
```

Haplotype	Carried lines	Reconstructed insertion
h1	A	ACGTTGCA (8 bp)
h2	A + B	ACGTGGTGCA (10 bp)
h3	A + S	ACATTGCA (8 bp)
h4	A + D	ACTGCA (6 bp)

One exact line; different complete assemblies

A always defines the same eight-base parent sequence at the same chr1 breakpoint. B inserts GG after base 4 of A; S changes base 3 from G to A; D deletes bases 3-4. The child lines state every difference explicitly. Thus sharing A does not mean the final insertion, after applying children, is identical in every carrier.

In exact output, each child likewise defines one shared sequence change at one parent-relative position. A record with CHROM=B can describe the next level. Carriers lacking a parent do not acquire its children.

Reconstruct from parent to child

For a represented haplotype region, start from its reference placement, resolve the carried top-level alleles, and recursively apply carried child operations within each parent. Interpret sibling positions on the unchanged parent coordinate axis; do not shift later coordinates by editing the string in an arbitrary order. Assembly contigs, query coordinates, and orientation locate the result in the original assembly.

The model also covers deletion differences

If a carrier deletes fewer bases than a deletion parent, retained bases can be expressed as a child insertion on that deleted interval. Full-locus copies can share a duplication template with nested differences. The hierarchy preserves these differences rather than reducing every carrier to a representative. [3]

05 / Shared sequence encoding

Nested records preserve differences between assembly sequences. Named sequence encoding complements that hierarchy by reusing shared bases; it does not change a variant's placement or substitute for its child records.

Reuse sequence without changing the placement

```
CHROM=chr1 POS=200 ID=C SVTYPE=INS SVLEN=6
SEQ=>A@3A2H4=2H>2ITT
```

After unescaping, SEQ is **>A:2H4=2H>2ITT**: copy A[2:6] (GTTG), then add TT, giving **GTTGTT**. The shared piece may reuse A's graph segments, while C remains attached at its own chr1 breakpoint. Sequence reuse and parent placement are separate relationships.

Encoding	Meaning
>target:body / <target:body	Forward / reverse-oriented named target. Reverse matches use reverse-complement bases.
nH / n= / nD	Target clipping / exact copy / skip target bases. H contributes no allele bases.
nXbases / nlbases / nSbases	Literal mismatch / insertion / soft-clipped query bases.
>nlbases	Unnamed literal piece, e.g. >2ITT adds TT.

Merged FORMAT can use the implicit row representative: >8= is an eight-base match. Current rGFA sequence verification rejects ambiguous M operations; use explicit =/X or literal sequence. Do not apply INFO/SEQ decoding rules blindly to FORMAT. [3, 4]

What lossless reconstruction requires

Retain SNPs, indels, SVs, complete dependency trees, and named sequence sources. Parent and child variants must be composed with their sample genotypes and placement/orientation metadata. A filtered subset, a missing catalog, or representative-only conversion cannot reproduce every assembly base.

Lossless is the representation goal, not a claim that every unmapped or filtered region is encoded. The per-sample checker `tools/check_vcf_lossless.py` compares reconstructed mapped regions with assembly sequence; it reports unmapped regions separately and currently requires plain SEQ. Validation is specific to the files and regions checked.

06 / Sequence and provenance

Identify the sequence mode

Producing stage	Where the allele sequence lives
Default final caller	<code>##minsetrefSequenceEncoding</code> declares <code>SEQ=plain</code> and <code>EXTENDGRAPHCIGARPayload=absent</code> . Bare alignment operations do not preserve every alternate base.
Caller with <code>--seqcompress</code>	<code>SEQ=omitted</code> , inline CIGAR payloads; <code>INFO/SEQ</code> is a dot. This is not directly sufficient for the current rGFA insertion verifier, which requires <code>SEQ</code> .
Merged insertion output	<code>SEQ</code> may be literal or a graph encoding that references other allele IDs or sequence catalogs. Resolve all named dependencies.

Preserve the original assemblies when per-observation bases are needed. In rGFA mode, literal insertion blocks and SNP/SUB alleles are checked against eligible carrier assembly intervals. An alignment alone is not proof of sequence identity. [2, 4]

Decode values after splitting the table

Core escapes are `@40` for `@`, `@3A` for colon, `@3B` for semicolon, `@2C` for comma, `@09` for tab, and `@0A` for newline. Some packed compatibility payloads also use `@7C` for `|`. Decode `@40` last. These are project conventions; retain them when passing grVCF between LinGraph tools.

Keep catalogs with their identity and placement metadata

Preserve `##contig` lengths, `##alternativeLocus` source metadata, and the original per-sample `##pseudoLinearMapping` entries. Required data can include the calling reference, indexed query assemblies, the alternative/novel catalog, and freshly lifted local-reference templates. Changing the backbone requires compatible calls; renaming CHROM does not lift variants.

`--local-path-fasta` translates local VCF targets onto included roots using source interval, orientation, and sequence identity. `-a` includes every nonempty supplied record as a complete additional path. Do not substitute `-a` for `--local-path-fasta` when using a catalog of duplicated local loci.

Full-locus duplication

`PACCLASS=fulllocusdup` and `ALTERNATIVECIGAR` describe copies aligned to source loci. The exporter can share duplication paths for overlapping source intervals while keeping mismatches and insertions as literal branches. Exact merging can represent full-locus copies through shared `DUP_` template paths and nested variants. In other merge paths, retention of full-locus duplication insertions is controlled by `--keep-full-locus-dup-insertions`. [3, 4]

Recover SNPs inside inserted sequence

```
python tools/extract_insertion_snps.py \  
-i cohort.sv.vcf cohort.indel.vcf \  
--assemblies query_paths.txt -o insertion.snp.vcf -t 8
```

The extractor uses saved observation alignments and recursively follows named mappings to destination coordinates, including reverse mappings. Supply indexed assemblies for bare X mismatches. Inputs must contain the needed insertion definitions; do not append complete VCF files by concatenating headers. [5]

07 / Build and use a graph

Commands here use the GitHub repository layout: pipeline programs under scripts/ and utilities under tools/. Run them from the repository root with your actual data paths.

```
python scripts/merged_vcf_to_gfa.py \
-v cohort.sv.vcf cohort.indel.vcf cohort.snp.vcf \
-q query_paths.txt --graph-folder graph \
--local-reference-templates local_reference_templates.fa \
--local-path-fasta graph/summary/alternatives.fasta \
-o cohort.gfa -t 8
```

Pass completed files for the same backbone; -v also accepts a single combined VCF or repeated inputs. query_paths.txt contains **NAME FASTA [FAI]**, with names matching sample columns and relative paths resolved against the list directory. Local-reference templates must have current placement metadata for the selected backbone.

Option / output	Current behavior
Default	--gfa-mode rgfa and --all: SNPs, substitutions, insertions, deletions, and all sizes. General --size-cutoff defaults to 0.
--svonly --svcutoff 20	Only non-SNP structural events meeting the size cutoff, including qualifying deletions.
--insertion-only [SIZE]	Restrict to insertions; a bare flag uses 50 bp.
Backbone	Default: graph/inputs/reference_alternatives_novels.fa. Use -r plus --reference-haplotype when selecting a plain assembly backbone.
--validate-only	Extract/check sequence and resolve topology without writing GFA. --bed-only is coordinate output, not equivalent validation.
--drop-unverified	Opt-in omission of unverifiable variants and their dependents, with unresolved reports. Default is to stop.
--partition-records N	Bound conversion work by groups of whole dependent variant trees for large cohorts.
Graph and sidecars	rGFA S/L/P records, stable LN/SN/SO/SR tags, *.variants.tsv and anchor/mapping/verification reports.

Insertion/SNP/SUB P paths contain allele cores; deletion paths use flanks to bypass the deleted interval. Exact mapped runs reuse segments, and the parent/reference route remains present. This updates the older insertion-only, full-parent-path description. [4]

Assembly walks are a separate output

```
python scripts/gfa_sample_gaf.py -g cohort.gfa \
-v cohort.sv.vcf cohort.indel.vcf cohort.snp.vcf \
-s samples/*/*.vcf -q query_paths.txt -o gaf -t 8
```

Use the same merged inputs in the same order as graph export. GAF generation splices carried variants using original sample mappings; walks break at unresolved intervals or topology gaps. It is not a guarantee of complete chromosome-length phased paths. [6]

08 / Export standard VCF

Standard VCF export is a reference-based projection, **not a lossless assembly representation**. It removes nested records rather than composing each carrier's child variants into its parent. In the worked example, retaining A alone cannot reproduce h2, h3, or h4. [7]

```
python tools/grvcf_to_vcf.py \  
  -i cohort.sv.vcf -r reference.fa \  
  -o cohort.sv.standard.vcf
```

Use the calling reference and adjacent .fai. Convert one input per invocation. Supply decoding catalogs with repeated **-a FASTA**; auxiliary catalogs do not add output reference contigs. Add **--normalize** for bcftools left alignment. BGZF .vcf.gz output and its CSI index also require bcftools.

Conversion result	What to expect
Explicit REF/ALT	SNPs, insertions, deletions, and replacements become base alleles with checked reference bases and indel padding.
Preserved record identity	IDs, sample names, genotypes, missingness, ploidy/phasing, QUAL, and FILTER are retained.
Standard annotations	Graph annotations are replaced with GT, AC, AN, AF, and NS. Genotypes support the chosen representative; member offsets and sequences are not expanded.
Graph-only records	Non-reference CHROM names beginning INS_, DEL_, SUB_, or DUP_ are removed and counted as nested_removed, not saved in the sidecar. Other non-reference rows go to the unplaced grVCF.
Sequence resolution	Named allele/FASTA dependencies and forward references are resolved. Missing bases, ambiguous M, conflicting IDs, or reference mismatches cause an error.

What to share with another user

Send the grVCF with its complete headers, the reference/backbone identity, required indexed sequence catalogs, and the producer version/options. Include original sample VCFs and assembly paths when downstream work needs sample walks or missing query bases. Keep the original complete grVCF even when sharing a standard VCF and its unplaced sidecar: removed nested variants are absent from both exported files.

Implementation sources

Revised 4 October 2026 from the updated README_grVCF_section.md and local implementations, including exact merging and nested-record removal. The primary design goal is lossless assembly representation; the exact-row guarantee is mode-dependent. Source links below identify the modules; use the version shipped with your calls.

[1] [VCF envelope reference](#)

[2] [Caller schema and mapping metadata](#)

[3] [Merged fields and nested events](#)

[4] [rGFA parser, allele metadata and coordinate resolver](#)

[5] [Insertion SNP extraction](#)

[6] [Sample GAF generation](#)

[7] [Standard VCF conversion](#)

[4] Also consult scripts/gfa_interval_metadata.py, gfa_stable_coords.py, gfa_interval_pipeline.py, and gfa_topology.py. Also see cohort_vcf_merge.py for exact-mode defaults and tools/check_vcf_lossless.py for reconstruction checks.